Jennifer Li

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Education

The University of Texas at Austin Austin, TX

Bachelor of Science in Electrical and Computer Engineering *May 2028*

Bachelor of Science in Mathematics

- GPA: 4.00/4.00
- Relevant Coursework: Circuit Theory, Embedded Systems, Linear Systems and Signals, Electrical & Computer Engineering Fundamentals, Linear Algebra, Differential Equations, Discrete Mathematics, Multivariable Calculus

Experience

Low Voltage Specialist Sep. 2025 – Present

Longhorn Racing Solar Vehicle Team – The University of Texas at Austin *Austin, TX*

- Test driver and one of three engineers validating the car's full power architecture through bench and on-car testing before competition; ensured it met Electrical and Battery Protection scrutineering standards at FSGP 2026
- Owned design and assembly of 6+ **MX150** and **Nano-Fit** harnesses for the motor controller, MPPTs, fans, lighting, and battery; sized wire gauge to load current and validated with crimp pull and continuity tests
- Reviewed schematics and layouts for 7+ boards in GitHub and in-person reviews, giving feedback on signal integrity, current paths, thermal margin, and part selection to optimize cost, manufacturability, and testability
- Mentor 3 new members on the car's LV power architecture, **KiCad**, **Git**, and clean schematic and layout practices

ECE Tutor Sep. 2026 – Present

Cockrell School of Engineering – The University of Texas at Austin *Austin, TX*

- Teach Intro to Electrical Engineering (ECE 402) and Intro to Computer Engineering (ECE 406) in two 3-hour weekly sessions available to 100+ students, covering topics on circuit analysis, assembly programming, etc
- Authored and disseminated problem sets, review packets, and concept guides (8+ so far) on difficult topics and recurring errors; guide students to reason through problems instead of giving answers, adapting to each skill level

Projects

Motor Controller & MPPT Interface PCBs Sep. 2025 – Jun. 2026

- Developed a 24V-to-12V power protection and thermal-management PCB keeping the motor controller and MPPTs below thermal shutdown; built 5 hot-swappable units with zero failures across all three FSGP 2026 race days
- Wrote cooling firmware in **embedded C** on an **STM32L4** with **FreeRTOS**: a **CAN** ISR feeds temperatures to a control task driving PWM fans, holding both subsystems under 55°C and forcing 100% duty after missed readings
- Root-caused dead fan outputs to a MOSFET gate threshold above the 3.3V drive and buck failures to a regulator run at its absolute-max input, using an oscilloscope; respun with logic-level FETs and higher-margin parts

Low-Voltage Carrier Board | *fabricated, in bring-up* Jun. 2026 – Present

- Engineered a **4-layer KiCad** PCB delivering 150W (12.5A at 12V) from a 120V, 5kWh pack to every LV system on the car; using a 92%-efficient isolated DC/DC to keep HV/LV grounds isolated, with ideal-diode source switching
- Built voltage, current, and power monitoring at four points using Hall-effect sensors chosen over shunt resistors in a cost, size, and availability trade study, reporting live system health over the vehicle's **CAN** bus for fault detection
- Designed an **STM32G4**-controlled precharge that gates the load until the bus charges, and a CC/CV buck-boost charger for the supplemental pack that throttles current when the main converter is loaded, plus LV cutoff

Addressable LED Strip Controller PCB Jun. 2026 – Present

- Built a USB-C powered LED controller around an **ESP32-S3** to drive custom color-changing and audio-reactive signs, running WS2812B strips from one cable carrying power and data; carried through a second revision
- Overcame USB current limits by implementing **USB-PD negotiation** for a 9V contract, bucked down to 5V, delivering roughly 2x the power to the LED rail versus stock 5V USB within the same cable current rating
- Engineered a multi-rail supply with input protection (fuse, TVS, reverse polarity): 9V-to-5V buck for the LED rail, 5V-to-3.3V LDO for the MCU, and a filtered analog rail isolating the microphone from digital switching noise

Technical Skills

Skills: Schematic capture, 4-layer PCB layout, power electronics, wire harness design, bring-up & debug, design review
Hardware & Lab: STM32 (L4/G4), CAN, I2C, PWM, ADC, oscilloscope, multimeter, bench power supply, soldering
Software & Tools: C (embedded), Python, Java, FreeRTOS, HAL, KiCad, Git/GitHub, STM32CubeIDE, VS Code